

ANALYZING THE BUSINESS USING ARCHITECTURES

*Applying End-to-End
Architecture for Value*





EXAMPLE: USING ARCHITECTURE TO ASSESS IMPACT ON THE ORGANIZATION

**THIS IS A SUPPLEMENT TO THE BPM AND
CASE MANAGEMENT GLOBAL SUMMIT HELD
ON JUNE 24, 2015.**

**THIS IS ASSOCIATED WITH THE
ARCHITECTURE WORKSHOP AND SHOULD BE
USED AS SUPPORTING THE LECTURE
MATERIALS.**



A typical problem:

If a company considers shifting markets and possibly products what is the impact on:

- Strategies,
- Business structure and
- Business operation/execution...running the business

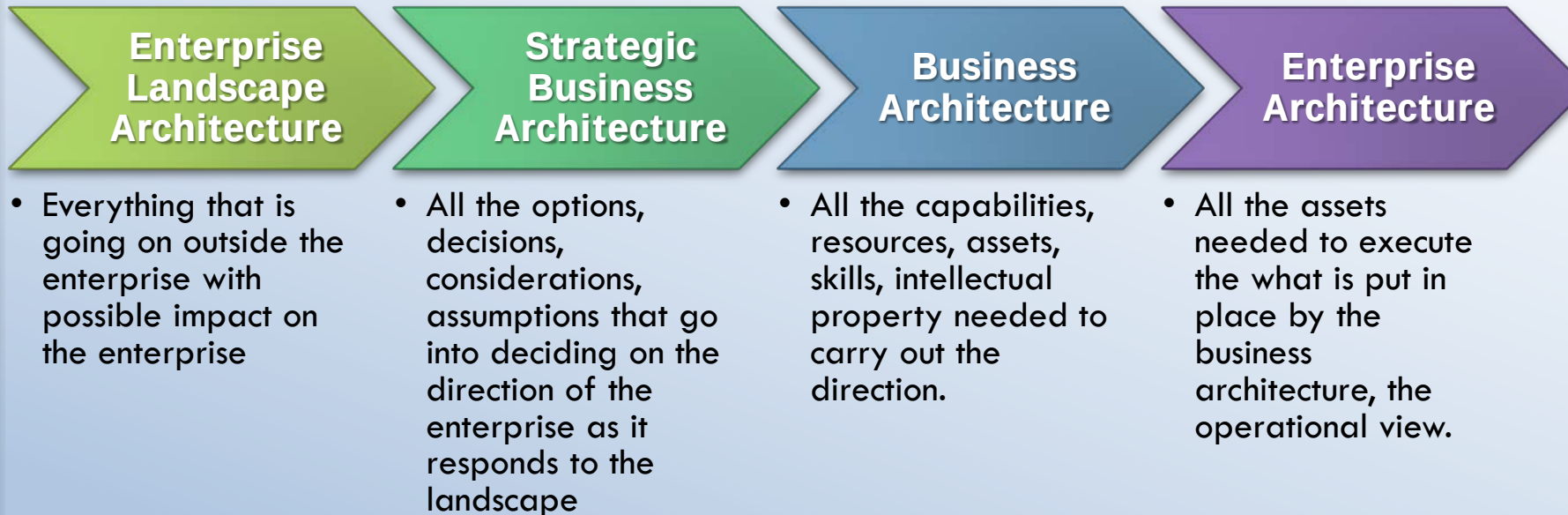
Example:

- An appliance company
- In geographic, demographic and product markets
- Performance is marginal

Typical analysis approach used in their business

- Fuzzy statement of intent for the analysis
- Multiple spreadsheets used for analysis
- Disparate data with unrelated context
- No point of integration
- Usually a point-in-time (snap-shot) analysis

The *End-to-End* relationship of architectures as used for Impact Analysis



By connecting the perspectives

There is more than one way to deal with an answer to this problem



What are the options?

- Business to Enterprise
- Strategy to Business
- External to Strategy
- Strategy to Business to Enterprise
- External to Strategy to Business
- External to Strategy to Business to Enterprise

This will depend on the scope as defined by management

Key analytics for architecture

Impact analysis for linkage between architecture perspectives

Comparative analytics to deal with consolidation, merger and acquisition

Ranking analytics similar to what is used for impact to identify options

Best fit for matching attributes along with components of the architecture

Affinity analysis to group things

Predictive analytics with clustering to identify hidden relationships for deployment



Setting up the solution approach using impact analysis:

**Management
direction:**

**Identify
eventual
impact on
business
execution**

- Link the four architecture perspectives together
 - External and Strategic
 - Strategic and Business
 - Business and Enterprise
- Assess impact perspective by perspective
- Identify the best and worst opportunities
- Start with either the external perspective or the Business/Enterprise pair
- Identify the most likely impacted enterprise execution components
- Suggest a plan to minimize the impact/turmoil

**Setting up the problem depends on where you start the effort,
in this case with external markets**

Each type of organization requires identification of those items used for analysis

NGO – Non-government organizations

Educational institutions

There are many different type of organizations

Charitable organizations

Trade associations

Religious bodies

Heritage organizations such as...Museums, Zoos, Conservancies, etc.

For example, governments may emphasize different components

External
environment
emphasizes

- Other country governments
- International relationships
- Political policies
- World and country social trends
- Trending legislation
- Economic and military alliances
- and so on

Strategic
architecture
deals with

- Missions and goals
- Ministries or agencies
- Commissions, working groups etc.
- Service delivery capabilities
- Value streams of services (in some cases products)

Business
architecture
focuses on

- Structure of organizations
- Related missions
- Legislated processes
- Funds management
- Purpose and role

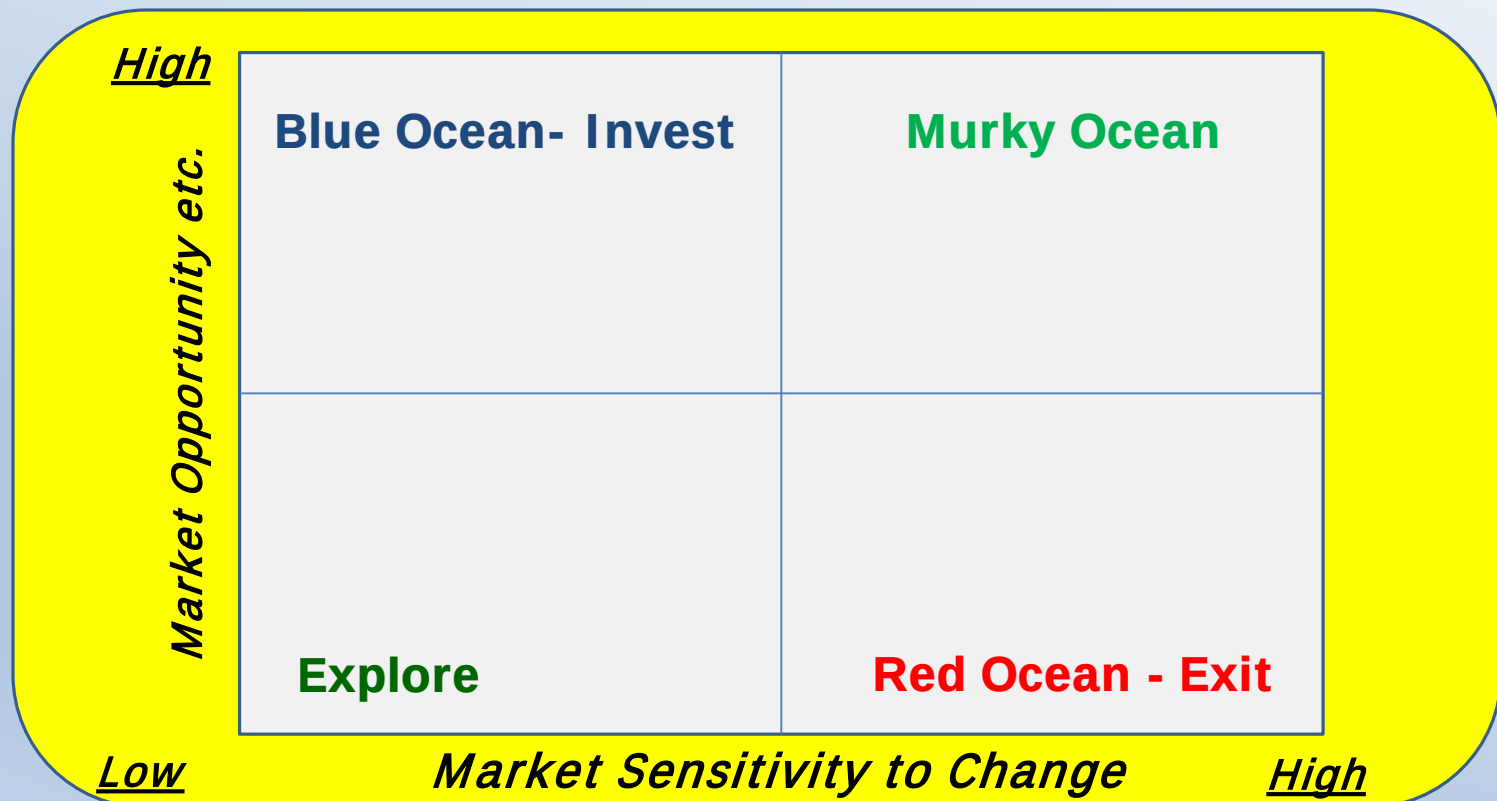
Enterprise
architecture is
focused on

- Interoperability of organizations, systems, database availability, weapons interoperability and so on
- Compliance to legislated mandates
- Process performance
- Delivery of services

Many components
are common
across organization
types.

Using the **Blue/Red** Ocean Idea

- A Blue ocean is the one in which there are few competitors and you have a hot product/service
- A Red ocean is where there are lots of products and lots of competitors and you are fighting for share
- There is a 4-Box way of looking at this



Organize the market information for consistent ranking

- Market Size – 5 is large (1 rank is a small market, sort ascending)
- Market share – Percentage (1 rank is small share, sort ascending)
- Number of products – largest/smallest (largest is too many, sort descending)
- Market potential – 5 is large (1 rank is too small, sort ascending)
- Market growth – 5 is large (1 rank is too small, sort ascending)
- Number of competitors – largest/smallest (largest is too many, sort descending)

Create a composite value such that:

1. The *smallest* composite number is the worst market to be in (Red)
2. The *largest* composite number is the best market to be in (Blue)

Collect some Marketing information

Market characteristics



KL	Name	Market Size (Int)	Market Share (Int)	Number of Products (Int)	Market Potential (Int)	Market Growth (Int)	Number of Competitors (Int)
	Demo - Age bracket	3	15	12	3	1	5
	Demo - Income	4	19	11	2	2	7
	Geo - Americas	5	31	41	1	1	5
	Geo - EU	3	5	9	2	1	7
	Geo - MENA	3	3	4	4	3	9
	Geo - SEA	4	7	4	3	2	11
	Prod - Air Conditioning	4	25	8	2	3	4
	Prod - Commercial water heaters	2	21	3	2	1	2
	Prod - Dishwashers	2	19	5	1	3	6
	Prod - Electronic cooking	2	10	2	3	2	3
	Prod - Gas cooking	4	26	6	4	3	5
	Prod - Heating	3	27	31	3	2	6
	Prod - Refrigeration	3	33	25	2	3	4
	Prod - Residentail water Heaters	4	21	21	2	1	3
	Prod- Home laundry	4	36	34	3	2	5

European Union (EU)

Middle East/ North Africa (MENA)

Southeast Asia (SEA)

Marketing provides:

- 6 characteristics that they use to evaluate a market that the company sells into.
- A starter list of markets

Tool used: Afterspyre

What does each characteristic look like ranked worst (number 1) to best (last)?

Run 1 -- A

ATTRIBUTES USED:

Market Size

INSTANCE RANK:

1 -- 1. -- Prod - Commercial water heaters
 1 -- 1. -- Prod - Electronic cooking
 1 -- 1. -- Prod - Dishwashers
 2 -- 2. -- Prod - Heating
 2 -- 2. -- Geo - EU
 2 -- 2. -- Prod - Refrigeration
 2 -- 2. -- Geo - MENA
 2 -- 2. -- Demo - Age bracket
 3 -- 3. -- Demo - Income
 3 -- 3. -- Geo - SEA
 3 -- 3. -- Prod - Air Conditioning
 3 -- 3. -- Prod - Gas cooking
 3 -- 3. -- Prod - Home laundry
 3 -- 3. -- Prod - Residentail water Heaters
 4 -- 4. -- Geo - Americas

Run 1 -- A

ATTRIBUTES USED:

Market Share

INSTANCE RANK:

1 -- 1. -- Geo - MENA
 2 -- 2. -- Geo - EU
 3 -- 3. -- Geo - SEA
 4 -- 4. -- Prod - Electronic cooking
 5 -- 5. -- Demo - Age bracket
 6 -- 6. -- Prod - Dishwashers
 6 -- 6. -- Demo - Income
 7 -- 7. -- Prod - Commercial water heaters
 7 -- 7. -- Prod - Residentail water Heaters
 8 -- 8. -- Prod - Air Conditioning
 9 -- 9. -- Prod - Gas cooking
 10 -- 10. -- Prod - Heating
 11 -- 11. -- Geo - Americas
 12 -- 12. -- Prod - Refrigeration
 13 -- 13. -- Prod - Home laundry

Run 1 -- A

ATTRIBUTES USED:

Number of Products

INSTANCE RANK:

1 -- 1. -- Geo - Americas
 2 -- 2. -- Prod - Home laundry
 3 -- 3. -- Prod - Heating
 4 -- 4. -- Prod - Refrigeration
 5 -- 5. -- Prod - Residentail water Heaters
 6 -- 6. -- Demo - Age bracket
 7 -- 7. -- Demo - Income
 8 -- 8. -- Geo - EU
 9 -- 9. -- Prod - Air Conditioning
 10 -- 10. -- Prod - Gas cooking
 11 -- 11. -- Prod - Dishwashers
 12 -- 12. -- Geo - MENA
 12 -- 12. -- Geo - SEA
 13 -- 13. -- Prod - Commercial water heaters
 14 -- 14. -- Prod - Electronic cooking

Run 1 -- A

ATTRIBUTES USED:

Market Potential

INSTANCE RANK:

1 -- 1. -- Prod - Dishwashers
 1 -- 1. -- Geo - Americas
 2 -- 2. -- Prod - Commercial water heaters
 2 -- 2. -- Geo - EU
 2 -- 2. -- Prod - Air Conditioning
 2 -- 2. -- Demo - Income
 2 -- 2. -- Prod - Refrigeration
 2 -- 2. -- Prod - Residentail water Heaters
 3 -- 3. -- Prod - Heating
 3 -- 3. -- Geo - SEA
 3 -- 3. -- Prod - Home laundry
 3 -- 3. -- Demo - Age bracket
 3 -- 3. -- Prod - Electronic cooking
 4 -- 4. -- Geo - MENA
 4 -- 4. -- Prod - Gas cooking

Run 1 -- A

ATTRIBUTES USED:

Market Growth

INSTANCE RANK:

1 -- 1. -- Demo - Age bracket
 1 -- 1. -- Prod - Commercial water heaters
 1 -- 1. -- Geo - Americas
 1 -- 1. -- Geo - EU
 1 -- 1. -- Prod - Residentail water Heaters
 2 -- 2. -- Geo - SEA
 2 -- 2. -- Prod - Home laundry
 2 -- 2. -- Prod - Electronic cooking
 2 -- 2. -- Demo - Income
 2 -- 2. -- Prod - Heating
 3 -- 3. -- Prod - Gas cooking
 3 -- 3. -- Prod - Air Conditioning
 3 -- 3. -- Prod - Dishwashers
 3 -- 3. -- Geo - MENA
 3 -- 3. -- Prod - Refrigeration

Run 1 -- A

ATTRIBUTES USED:

Number of Competitors

INSTANCE RANK:

1 -- 1. -- Geo - SEA
 2 -- 2. -- Geo - MENA
 3 -- 3. -- Demo - Income
 3 -- 3. -- Geo - EU
 4 -- 4. -- Prod - Dishwashers
 4 -- 4. -- Prod - Heating
 5 -- 5. -- Prod - Gas cooking
 5 -- 5. -- Demo - Age bracket
 5 -- 5. -- Prod - Home laundry
 5 -- 5. -- Geo - Americas
 6 -- 6. -- Prod - Refrigeration
 6 -- 6. -- Prod - Air Conditioning
 7 -- 7. -- Prod - Electronic cooking
 7 -- 7. -- Prod - Residentail water Heaters
 8 -- 8. -- Prod - Commercial water heaters

The rankings are combined into a composite rank factoring in all 6 characteristics

Market characteristics

Run 1 -- A | Run 2 -- A | Run 3 -- A | F

ATTRIBUTES USED:
Market Size
Market Share
Number of Products
Market Potential
Market Growth
Number of Competitors

INSTANCE RANK:
1 -- 3. -- Geo - EU
2 -- 3.67 -- Demo - Age bracket
3 -- 3.83 -- Demo - Income
3 -- 3.83 -- Geo - Americas
4 -- 4. -- Prod - Heating
4 -- 4. -- Geo - MENA
4 -- 4. -- Geo - SEA
5 -- 4.17 -- Prod - Residentail water Heaters
6 -- 4.33 -- Prod - Dishwshers
7 -- 4.67 -- Prod- Home laundry
8 -- 4.83 -- Prod - Refrigeration
9 -- 5.17 -- Prod - Air Conditioning
9 -- 5.17 -- Prod - Electronic cooking
10 -- 5.33 -- Prod - Commercial water heaters
11 -- 5.67 -- Prod - Gas cooking

Worst market to enter or be in

This composite analysis was created in about 30 seconds

Best market to enter or be in

Using PESTLE for market sensitivity, the PESTLE factor impact details are identified with a set of matrices

Model Name: Economic Impact on Markets [From Dimension: EXTERNAL ARCHITECTURE -- To Dimension: EXTERNAL ARCHITECTURE]

All
Expand Contract Class Relations: 115 - Density: 40.36%

From	To	Aggregate Supply (Economic)	Business Formations	Household Formations	Imports (Economic)	Price Level (Economic)	Disposable Income	Inflation (Economic)	Consumption (Economic)	Government Spending	Exports (Economic)
Demo - Age brack											
Geo - Americas (M)											
Prod - Heating (M)											
Prod - Refrigeratio											
Demo - Income (M)											

Model Name: Market Impact by Regulation [From Dimension: EXTERNAL ARCHITECTURE -- To Dimension: EXTERNAL ARCHITECTURE]

All
Expand Contract Class Relations: 58 - Density: 27.62%

From	To	CPSC - Consumer Product Safety	EPA - Environmental	FTC - Federal Trade Commission	NLRB - National Labor Relations	OSHA - Occupational	ICC - Interstate Commerce	EEOC - Equal Employment
Demo - Age brack								
Geo - Americas (M)								
Geo - MENA (Mar)								
Geo - SEA (Marke								
Prod - Electronic c								
Prod - Gas cookin								
Prod - Home laund								
Prod - Dishwashers								
Prod - Air Conditi								

Model Name: Market Impact by Regulation [From Dimension: EXTERNAL ARCHITECTURE -- To Dimension: EXTERNAL ARCHITECTURE]

All
Expand Contract Class Relations: 58 - Density: 27.62%

From	To	CPSC - Consumer Product Safety	EPA - Environmental	FTC - Federal Trade Commission	NLRB - National Labor Relations	OSHA - Occupational	ICC - Interstate Commerce	EEOC - Equal Employment	FCC - Federal Communication
Demo - Age brack									
Geo - Americas (M)									
Prod - Dish									
Prod - Ga									
Prod - Re									
Prod - Ho									
Prod - Ho									
Prod - Ho									
Prod - Ho									
Prod - Ho									

Model Name: Market Impact by Social [From Dimension: EXTERNAL ARCHITECTURE -- To Dimension: EXTERNAL ARCHITECTURE]

All
Expand Contract Class Relations: 99 - Density: 41.33%

From	To	Agile Marketing (Social)	Currencies of change (Social)	Geo - Fencing (Social)	Geo - Targeting (Social)	Social Video (Social)	Social Commerce (Social)	Integrated Advertising	Social Networking	Graying society (Social)	More Singles/never marrieds	Niche Networks (Social)
Demo - Age brack												
Geo - Americas (M)												
Prod - Dish												
Geo - EU												
Geo - ME												
Geo - SE												
Prod - Ele												
Prod - Air												
Prod - He												

Model Name: Technology Impacts on Markets [From Dimension: EXTERNAL ARCHITECTURE -- To Dimension: EXTERNAL ARCHITECTURE]

All
Expand Contract Class Relations: 84 - Density: 35.0%

From	To	3D Printing (Technologies)	Autonomous Vehicles	Biometric (Technologies)	Industrial Robotics	Nanotechnology (Technologies)	Sensor Technology	Social media (Technologies)	Mobile Computing (Technologies)	Internet of Things (Technologies)	Cyber Risk (Technologies)	Cloud Comput
Demo - Age brack												
Demo - Income (M)												
Geo - Americas (M)												
Prod - Electronic c												
Prod - Gas cookin												
Prod - Home laund												
Prod - Dishwashers												
Prod - Heating (M)												
Prod - Residential												
Geo - EU (Markets)												
Geo - MENA (Mar)												
Geo - SEA (Marke												
Prod - Electronic c												
Prod - Gas cookin												
Prod - Refrigeratio												
Prod - Home laund												

5 PESTLE factors are used here. The rows are the markets and the columns are the details of the PESTLE factor

Like with the market characteristics we need a composite of the 5 PESTLE components

Markets --> Legislative
Matrix) Economic Impact on Markets
Markets --> Economic
Matrix) Market Impact by Social
Markets --> Social
Matrix) Technology Impacts on Markets
Markets --> Technologies

INSTANCE RANK:

1	--	1.25	--	Geo - Americas
2	--	3.25	--	Geo - SEA
2	--	3.25	--	Geo - MENA
2	--	3.25	--	Demo - Income
3	--	3.5	--	Prod - Air Conditioning
3	--	3.5	--	Geo - EU
4	--	3.75	--	Prod - Electronic cooking
4	--	3.75	--	Demo - Age bracket
5	--	4.5	--	Prod - Refrigeration
5	--	4.5	--	Prod - Commercial water heaters
5	--	4.5	--	Prod - Heating
5	--	4.5	--	Prod - Gas cooking
6	--	4.75	--	Prod - Home laundry
7	--	5.	--	Prod - Residential water Heaters
8	--	5.25	--	Prod - Dishwashers

*Least
Sensitive*

The composite ranking looks like this where the lowest value is the least sensitive market and the highest is the most sensitive

*Most
Sensitive*

Tool used: Afterspyre

The idea is to put both rankings together and determine what the company should focus on

Run 1 -- A | Run 2 -- A | Run 3 -- A | F

ATTRIBUTES USED:

Market Size
Market Share
Number of Products
Market Potential
Market Growth
Number of Competitors

INSTANCE RANK:

1 -- 3. -- Geo - EU
2 -- 3.67 -- Demo - Age bracket
3 -- 3.83 -- Demo - Income
3 -- 3.83 -- Geo - Americas
4 -- 4. -- Prod - Heating
4 -- 4. -- Geo - MENA
4 -- 4. -- Geo - SEA
5 -- 4.17 -- Prod - Residentail water Heaters
6 -- 4.33 -- Prod - Dishwshers
7 -- 4.67 -- Prod- Home laundry
8 -- 4.83 -- Prod - Refrigeration
9 -- 5.17 -- Prod - Air Conditioning
9 -- 5.17 -- Prod - Electronic cooking
10 -- 5.33 -- Prod - Commercial water heaters
11 -- 5.67 -- Prod - Gas cooking

Worst Market Opportunity

Least impacted by Pestle

Most impacted

Best Market Opportunity

Markets --> Legislative
Matrix) Economic Impact on Markets
Markets --> Economic
Matrix) Market Impact by Social
Markets --> Social
Matrix) Technology Impacts on Markets
Markets --> Technologies

INSTANCE RANK:

1 -- 1.25 -- Geo - Americas
2 -- 3.25 -- Geo - SEA
2 -- 3.25 -- Geo - MENA
2 -- 3.25 -- Demo - Income
3 -- 3.5 -- Prod - Air Conditioning
3 -- 3.5 -- Geo - EU
4 -- 3.75 -- Prod - Electronic cooking
4 -- 3.75 -- Demo - Age bracket
5 -- 4.5 -- Prod - Refrigeration
5 -- 4.5 -- Prod - Commercial water heaters
5 -- 4.5 -- Prod - Heating
5 -- 4.5 -- Prod - Gas cooking
6 -- 4.75 -- Prod- Home laundry
7 -- 5. -- Prod - Residentail water Heaters
8 -- 5.25 -- Prod - Dishwshers

Use the data to create a 4 – box starting with arranging the data in pairs based on the market name

Market Factors

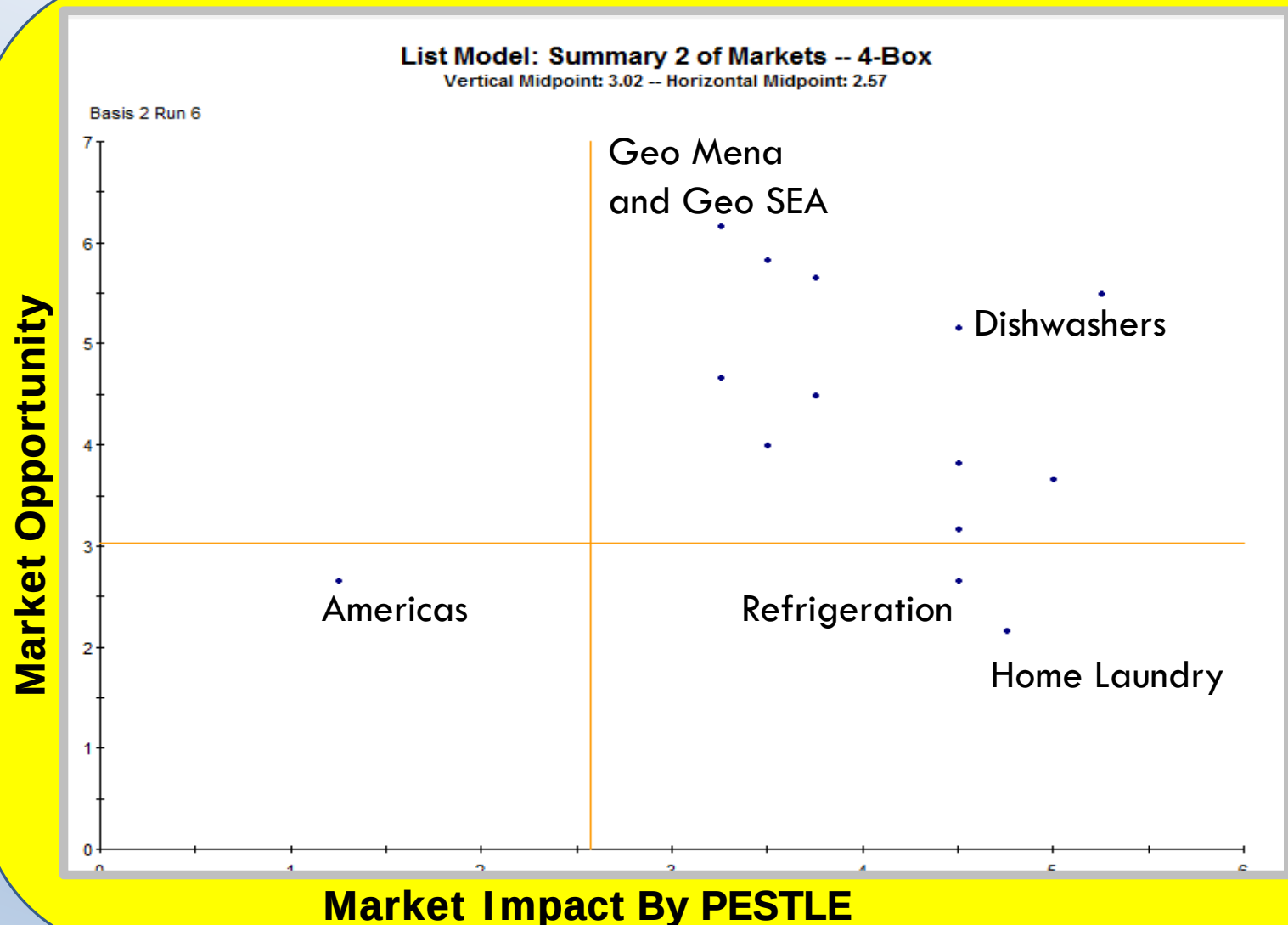
Market Landscape (sensitivity)

No.	15 Relations	Basis 2 Run	Context 2
1	Demo - Age bracket	4.5	3.75
2	Demo - Income	4.67	3.25
3	Geo - Americas	2.67	1.25
4	Geo - EU	5.83	3.5
5	Geo - MENA	6.17	3.25
6	Geo - SEA	6.17	3.25
7	Prod - Air Conditioning	4.	3.5
8	Prod - Commercial water he	5.17	4.5
9	Prod - Dishwashers	5.5	5.25
10	Prod - Electronic cooking	5.67	3.75
11	Prod - Gas cooking	3.83	4.5
12	Prod - Heating	3.17	4.5
13	Prod - Refrigeration	2.67	4.5
14	Prod - Residentail water He	3.67	5.
15	Prod- Home laundry	2.17	4.75

From this data create a logistic regression – a 4 box, we get the following:

Next Analytic creates the 4 – box

Where are most of their markets?



What do we know from the 4 – box and rankings?

- *Observations:*

- The large number of markets in the ‘murky’ space indicate further refinement may be needed
- Segmenting the market needs better input from marketing
- Consider separating the types of markets
- The two markets in the **Red Ocean** should be assessed
 - ✓ Home laundry (may want to look at details of these markets)
 - ✓ Refrigeration
- The Americas seem like a better market to invest in
 - ✓ Least impact but
 - ✓ More difficult (explore why by looking at market characteristics)

Quantitative considerations

- The view thus far is a 'snapshot' of the markets at a point in time
- Market trends over time should be analyzed
- Probability factors can be used to indicate the confidence in the relationships and the results
- Income statement and balance sheet values can be integrated with the conclusions
 - What does it cost to increase market share in a competitive market?
 - Do we have the investment to go into a new market?

Market trends over time

- What is the direction of factors for each market?
- Are they increasing, decreasing, stable
- What factors should be viewed together e.g.
 - Market size, market share, and number of products or services sold, e.g.
 - ✓ Sales may be up but the market size is increasing and your share is going down
 - ✓ Your share is going up, product and service sales are the same means the market may be stagnating
 - Rate of change of market size increasing versus your share, probably decreasing if sales are not up to match the 'ramp up' of a market
 - These factors may be included in the rankings using options in the software that create the composite
 - How these factors work together is the purview of marketing and strategic planning.

Next set of analytics are to assess impact:

External architecture is linked with *strategic* architecture

This is done in 2 parts:

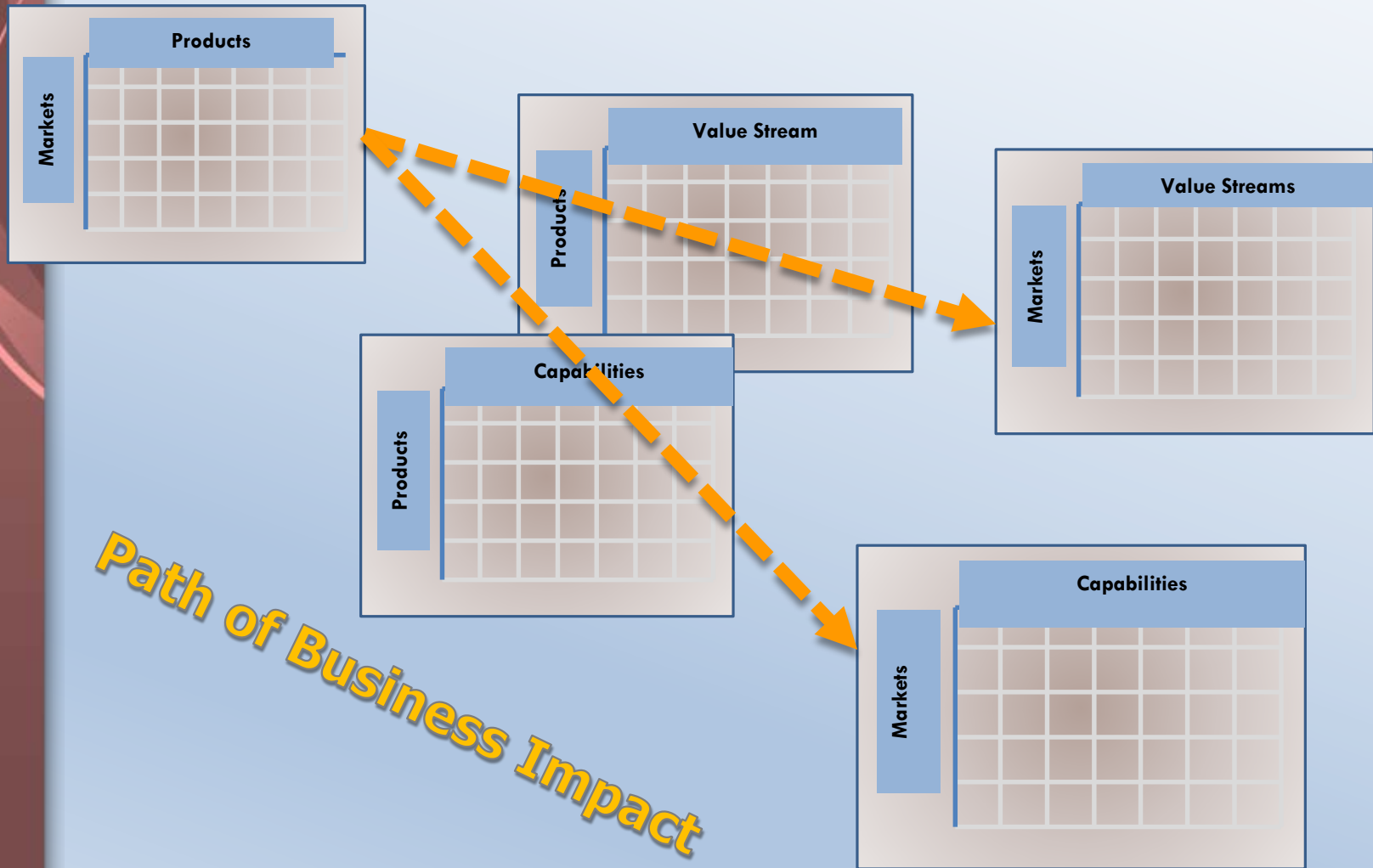
- Strategic planning and marketing have the list of products the company sells. This is part of the strategic architecture
- Connect products to markets using management, marketing and strategic planning input. This is done in a workshop.

All
☐ Expand ☒ Contract Class Relations: 37 -- Density: 27.41%

From	To	Air Conditioning (Appliances)	Home Laundry (Appliances)	Refrigeration (Appliances)	Gas Cooking (Appliances)	Residential Water Heaters	Commercial Water Heaters	Electronic Cooking	Dish Washers (Appliances)	Heating (Appliances)
Demo - Age brack		●	●						●	
Demo - Income (M				●	●				●	●
Geo - Americas (M		●	●		●	●	●		●	●
Geo - EU (Markets		●			●	●		●		●
Geo - MENA (Marl		●		●		●		●		
Geo - SEA (Marke		●	●			●		●		
Prod - Electronic c								●		
Prod - Gas cookin					●					
Prod - Refrigeratio				●						
Prod - Home laund			●							
Prod - Air Condition		●								
Prod - Heating (Ma						●				●
Prod - Residentail						●				
Prod - Commercial							●			

Infer the impact of the *strategic* architecture to the *business* architecture

Use Value Streams and Capabilities



Create two matrices that relate products to capabilities and value streams

All
☐ Expand ☒ Contract Class Relations: 42 -- Density: 46.67%

From	To	Customer Relationship Management (Company A)	Customer Service Management (Company A)	Manage Product Acquisition (Company A)	Product Design Management (Company A)	Product Engineering (Company A)	Manufacturing Management (Company A)	Personnel management (Company A)	Vendor Management (Company A)	Financial Management (Company A)
Refrigeration (Appliances)		●	●	●	●	●	●	●		
Dish Washers (Appliances)									●	●
Home Laundry (Appliances)		●	●	●					●	●
Residential Water Heaters (Appliances)										●
Commercial Water Heaters (Appliances)		●	●	●						●
Electronic Cooking (Appliances)		●	●	●	●	●	●	●		
Air Conditioning (Appliances)										
Heating (Appliances)										
Gas Cooking (Appliances)										

Capabilities

All
☐ Expand ☒ Contract Class Relations: 33 -- Density: 36.67%

From	To	Maintain Products (Company A)	Service Customers (Company A)	Supply Parts (Company A)	Ship to Customer (Company A)	Manage Information (Company A)	Select Suppliers (Company A)	Build Products (Company A)	Manage Supply Chain (Company A)
Refrigeration (Appliances)		●	●	●	●	●	●	●	
Dish Washers (Appliances)									●
Home Laundry (Appliances)									●
Residential Water Heaters (Appliances)									●
Commercial Water Heaters (Appliances)									●
Electronic Cooking (Appliances)		●	●	●	●	●	●	●	
Air Conditioning (Appliances)		●	●	●	●	●	●	●	
Heating (Appliances)		●	●	●	●	●	●	●	
Gas Cooking (Appliances)									●

Value Streams

Infer the relationship between the markets and capabilities

Notice the top set of rows for the *Demographic* and *Geographic* markets – they are too dense (all cells with dots, no discrimination) for any conclusion. Focus on the Product Markets

All
☐ Expand ☒ Contract Class Relations: 98 -- Density: 65.33%

From	To	Customer Relationship Management	Customer Service Management	Financial Management	Manage Product Acquisition	Manufacturing Management	Personnel management	Product Design Management	Product Engineering	Vendor Management
Demo - Age bracket		•	•	•	•	•	•	•	•	•
Demo - Income (M		•	•	•	•	•	•	•	•	•
Geo - Americas (M		•	•	•	•	•	•	•	•	•
Geo - EU (Markets		•	•	•	•	•	•	•	•	•
Geo - MENA (Marl		•	•	•	•	•	•	•	•	•
Geo - SEA (Marke		•	•	•	•	•	•	•	•	•
Prod - Air Condition			•			•	•	•	•	
Prod - Commercial		•	•	•	•					•
Prod - Dishwashers				•						•
Prod - Electronic c		•	•		•	•	•	•	•	
Prod - Gas cookin		•	•	•	•					•
Prod - Heating (Ma				•		•	•	•	•	•
Prod - Refrigeratio		•	•		•	•	•	•	•	
Prod - Residential				•						•

Look at the cutoff points for both markets and capabilities, what do you see?

Markets		Capabilities	
Affinity Workspace			
Row Frequencies		Column Frequencies	
Instance Name	Frequency	Instance Name	Frequency
Geo - SEA	9	Financial Management	12
Geo - MENA	9	Vendor Management	12
Geo - EU	9	Customer Service Management	12
Geo - Americas	9	Manage Product Acquisition	11
Demo - Income	9	Customer Relationship Management	11
Demo - Age bracket	9	Personnel management	10
Prod - Refrigeration	7	Manufacturing Management	10
Prod - Electronic cooking	7	Product Engineering Management	10
Prod - Heating	6	Product Design Management	10
Prod - Commercial water heaters	5		
Prod - Air Conditioning	5		
Prod - Home laundry	5		
Prod - Gas cooking	5		
Prod - Residential water Heaters	2		
Prod - Dishwashers	2		

Do the same for value streams

Notice the density of the top set of rows for the Demographic and Geographic markets – Also too dense

All
☐ Expand ☒ Contract Class Relations: 82 -- Density: 54.67%

From	To	Build Products (Company A Val. Stream)	Maintain Products (Company A Val. Stream)	Manage Information (Company A Val. Stream)	Manage Supply Chain (Company A Val. Stream)	Select Suppliers (Company A Val. Stream)	Service Customers (Company A Val. Stream)	Ship to Customer (Company A Val. Stream)	Supply Parts (Company A Val. Stream)
Demo - Age brack		●	●	●	●	●	●	●	●
Demo - Income (M		●	●	●	●	●	●	●	●
Geo - Americas (M		●	●	●	●	●	●	●	●
Geo - EU (Markets		●	●	●	●	●	●	●	●
Geo - MENA (Marl		●	●	●	●	●	●	●	●
Geo - SEA (Marke		●	●	●	●	●	●	●	●
Prod - Air Condition		●	●	●		●	●	●	●
Prod - Commercial					●				
Prod - Dishwashers					●				
Prod - Electronic c		●	●	●		●	●	●	●
Prod - Gas cookin					●				
Prod - Heating (Ma		●	●	●	●	●	●	●	●
Prod - Refrigeratio		●	●	●		●	●	●	●
Prod - Residentail					●				

Another set of cutoff points.

Markets		Value Streams	
Affinity Workspace			
Row Frequencies		Column Frequencies	
Instance Name	Frequency	Instance Name	Frequency
Prod - Heating	8	Manage Supply Chain	12
Geo - SEA	8	Build Products	10
Geo - MENA	8	Select Suppliers	10
Geo - EU	8	Manage Information	10
Geo - Americas	8	Ship to Customer	10
Demo - Income	8	Supply Parts	10
Demo - Age bracket	8	Service Customers	10
Prod - Air Conditioning	7	Maintain Products	10
Prod - Refrigeration	7		
Prod - Electronic cooking	7		
d - Commercial water heaters	1		
d - Residentail water Heaters	1		
Prod - Dishwshers	1		
Prod- Home laundry	1		
Prod - Gas cooking	1		

What do we know from the first level inference?

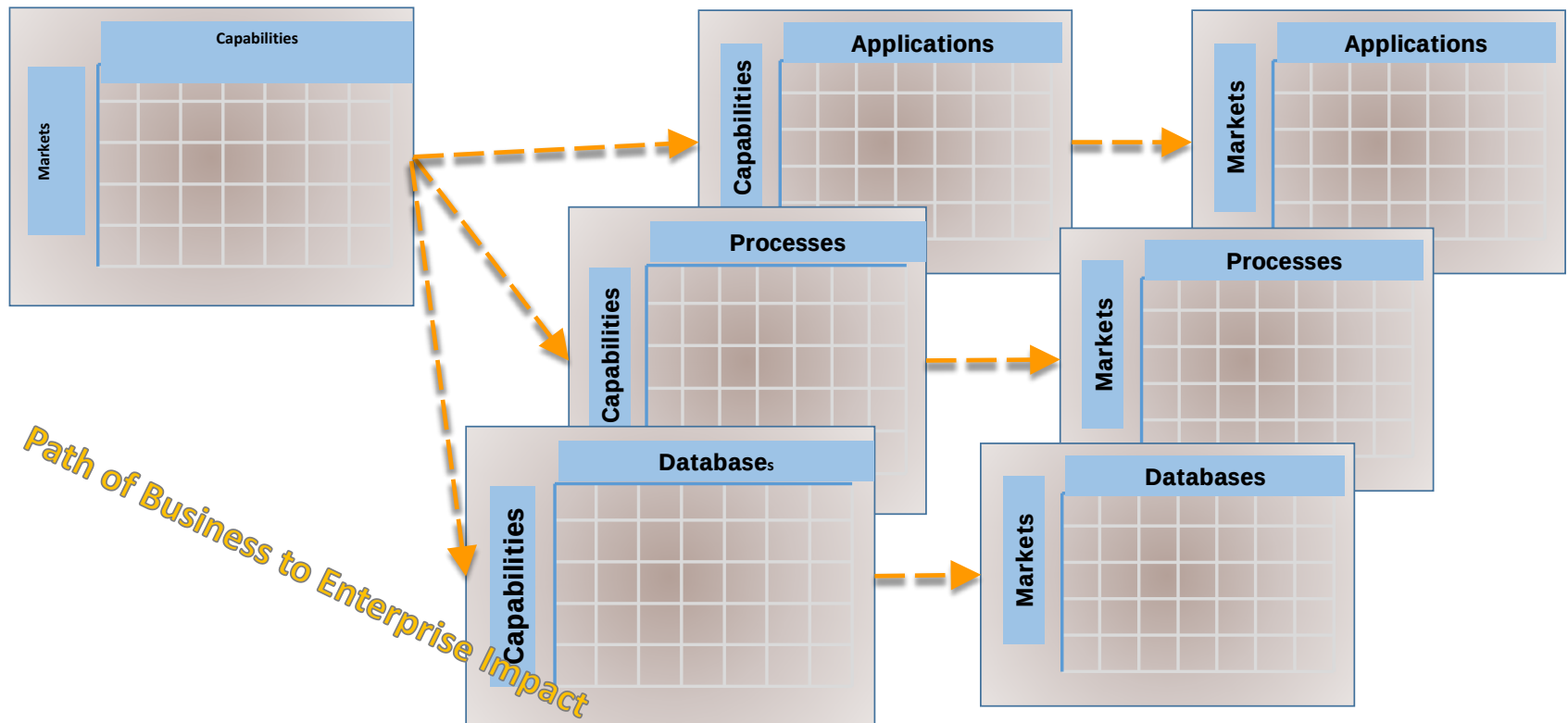
- *Observation:*
 - Very dense matrix in the first two types of market perspectives, Geographic and Demographic
- **Conclusions:**
 - No market discrimination from a very dense matrix
 - Not enough detail segmentation in this market data
- **Suggestions:**
 - Reduce the density of the matrix by separating out the three market perspectives
 - Use more detailed segmentation data
 - Create 3 analysis paths one for each market perspective:
 - ✓ Geographic
 - ✓ Demographic
 - ✓ Product

Now connect the market perspectives from *business* architecture to 3 of the *enterprise* architecture components

Use Applications, Databases and Processes

Key Linkage of Markets to Applications, Databases and Processes

Choose one path to work with, in this case Processes



The inferred matrix result:

Model Name: Company A Inferred Market to Processes [From Dimension: EXTERNAL ARCHITECTURE -- To Dimension: FUNCTIONS]

All

☐ Expand ☒ Contract Class Relations: 93 -- Density: 18.24%

From	To	Build Appliances (DAC Core)	Develop Human Resources (DAC Core)	Develop New Products (DAC Core)	Manage Quality (DAC Core)	Manage Supply Chain (DAC Core)	Market Appliances (DAC Core)	Report on Financial Results (DAC Core)	Research and Develop (DAC Core)
Demo - Age brack		●	●	●	●	●	●	●	●
Demo - Income (M		●	●	●	●	●	●	●	●
Geo - Americas (M		●	●	●	●	●	●	●	●
Geo - EU (Market		●	●	●	●	●	●	●	●
Geo - MENA (Marl		●	●	●	●	●	●	●	●
Geo - SEA (Marke		●	●	●	●	●	●	●	●
Prod - Air Condition		●	●	●	●	●			●
Prod - Commercial		●				●	●	●	
Prod - Dishwashers		●				●		●	
Prod - Electronic c		●	●	●	●	●	●		●
Prod - Gas cookin		●				●	●	●	
Prod - Heating (Ma		●	●	●	●	●		●	●
Prod - Refrigeratio		●	●	●	●	●	●		●
Prod - Residentail		●				●		●	

Markets > DAC Core Processes

Look at the top set of markets and processes:

Affinity Workspace

Row Frequencies	
Instance Name	Frequency
Geo - SEA	8
Geo - MENA	8
Geo - EU	8
Geo - Americas	8
Demo - Income	8
Demo - Age bracket	8
Prod - Heating	7
Prod - Refrigeration	7
Prod - Electronic cooking	7
Prod - Air Conditioning	6
d - Commercial water heaters	4
Prod- Home laundry	4
Prod - Gas cooking	4
d - Residentail water Heaters	3
Prod - Dishwshers	3

Column Frequencies	
Instance Name	Frequency
Manage Supply Chain	15
Build Appliances	15
Report on Financial Results	12
Market Appliances	11
earch and Develop Products	10
Develop Human Resources	10
Manage Quality	10
Develop New Products	10

Observation: *Manage Supply Chain* is an important process that supports *Building Appliances*. If management changes markets, these processes will be impacted.

European Union (EU)
Middle East/ North Africa (MENA)
Southeast Asia (SEA)

Looking now at market and applications

Affinity Workspace

Row Frequencies		Column Frequencies	
Instance Name	Frequency	Instance Name	Frequency
Geo - SEA	11	Supply Chain Management	13
Geo - MENA	11	Financial Analysis	12
Geo - EU	11	Order Fulfillment	12
Geo - Americas	11	Financial Reporting	12
Demo - Income	11	Order Entry	12
Demo - Age bracket	11	Customer Analysis	12
d - Commercial water heaters	8	Accounting	12
Prod - Home laundry	8	Shipping and Delivery System	11
Prod - Refrigeration	8	Employee Management	10
Prod - Gas cooking	8	Inventory management	10
Prod - Electronic cooking	8	Production Control	10
Prod - Heating	7		
Prod - Air Conditioning	7		
d - Residentail water Heaters	3		
Prod - Dishwshers	3		

European Union (EU)
 Middle East/ North Africa (MENA)
 Southeast Asia (SEA)

Observation: *Supply Chain Management* supports the Supply Chain process and will also be impacted.

Finally markets and databases

Affinity Workspace

Row Frequencies	
Instance Name	Frequency
Geo - SEA	9
Geo - MENA	9
Geo - EU	9
Geo - Americas	9
Demo - Income	9
Demo - Age bracket	9
Prod - Heating	7
Prod - Air Conditioning	7
Prod - Refrigeration	7
Prod - Electronic cooking	7
d - Commercial water heaters	6
Prod - Home laundry	6
Prod - Gas cooking	6
d - Residential water Heaters	3
Prod - Dishwashers	3

Column Frequencies	
Instance Name	Frequency
Supplier	15
Product	13
Finance	12
Sales	12
Accounts	12
Customer	12
Inventory	10
Parts	10
Employee	10

Observation: *Supplier and Product* supports the Supply Chain process and will also be impacted. Hence, market changes will impact the Supply Chain and related capabilities.

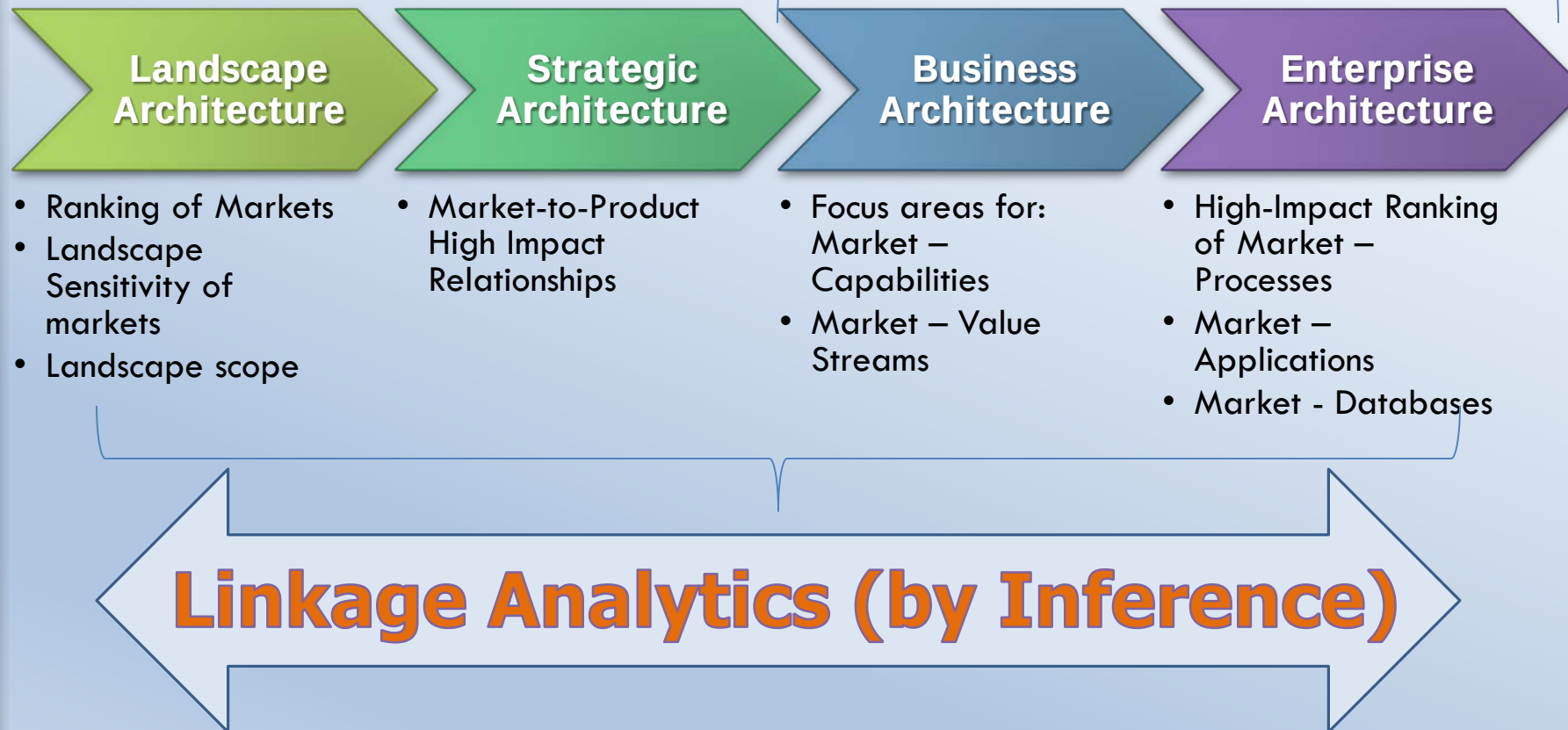
European Union (EU)
Middle East/ North Africa (MENA)
Southeast Asia (SEA)

What do we know from the second level of inference?

- *Observations:*
 - Looking across all 3 rankings, markets that stand out are:
 - ✓ Middle East/ North Africa (MENA)
 - ✓ Southeast Asia (SEA)
 - ✓ European Union (EU)
 - ✓ Americas
 - These are geographic markets
 - These geographic markets have the most indicated impact on the three Enterprise Architecture components
 - In fact the geographic markets and demographic markets show the most influence supporting the indication that they should be part of a separate analysis.
 - The 3 rankings also point to the supply chain and related data and applications as the number one impacted capability
 - As this is a manufacturing organization, it may make sense

The overall End-to-End result focusing on Impact Analysis outcomes ...

Most Organizations do some analysis here, usually requirements analysis



A few final topics...

**Understand
the problem
and the
management
need:**

- Sometimes change comes from two directions and you need to deal with it.
- It is useful to infer from Enterprise Architecture to Strategic Architecture
- Quantitative techniques augment the qualitative techniques
- There is much more than impact analysis regarding architecture analytics

There is an issue with the analysis that relates to markets

- After the first inference the top rows of the result have all cells filled
- This is an indication that there are overlaps in the segmentation approach
- A review of the market segmentation would show that hits can be in more than one category of market:
 - A product may also be in a geographic or demographic market
 - The action at that point in the analysis is to split the market segmentation into 3 parts
 - The marketing function in an organization should pick this issue up quickly
 - The next slide shows a suggested segmentation

A suggested market segmentation approach

Demographics

Age:

20 – 29

30 – 39

40 – 49

50 – 59

60 +

Income

30K

31 – 60K

61 – 90K

91K+

Social Media Use

Family size

Marital status

Etc.

Geographical

Region:

North America

South America

Central America

China

Asia – Pacific

Middle East

Northern Africa

Southern Africa

EU West

EU East

Also by:

Country

Statistical area

Population

Etc.

Products

Commercial water heaters

Electronic cooking

Dishwashers

Heating

Refrigeration

Air conditioning

Gas cooking

Home laundry

Residential water heaters

This will result in a more detailed analysis

Each market segmentation would be analyzed like the overall group

A ranking would be developed by segment

The final rank would be a composite off the 3 segmentation approaches

Marketing might suggest more segmentation approaches

Management might suggest a different set of things they want impact information on

This would be an iterative process

The Business Architecture 'Squeeze'

The organization changes the technology as part of Enterprise Architecture

Management changes strategies that impact the business

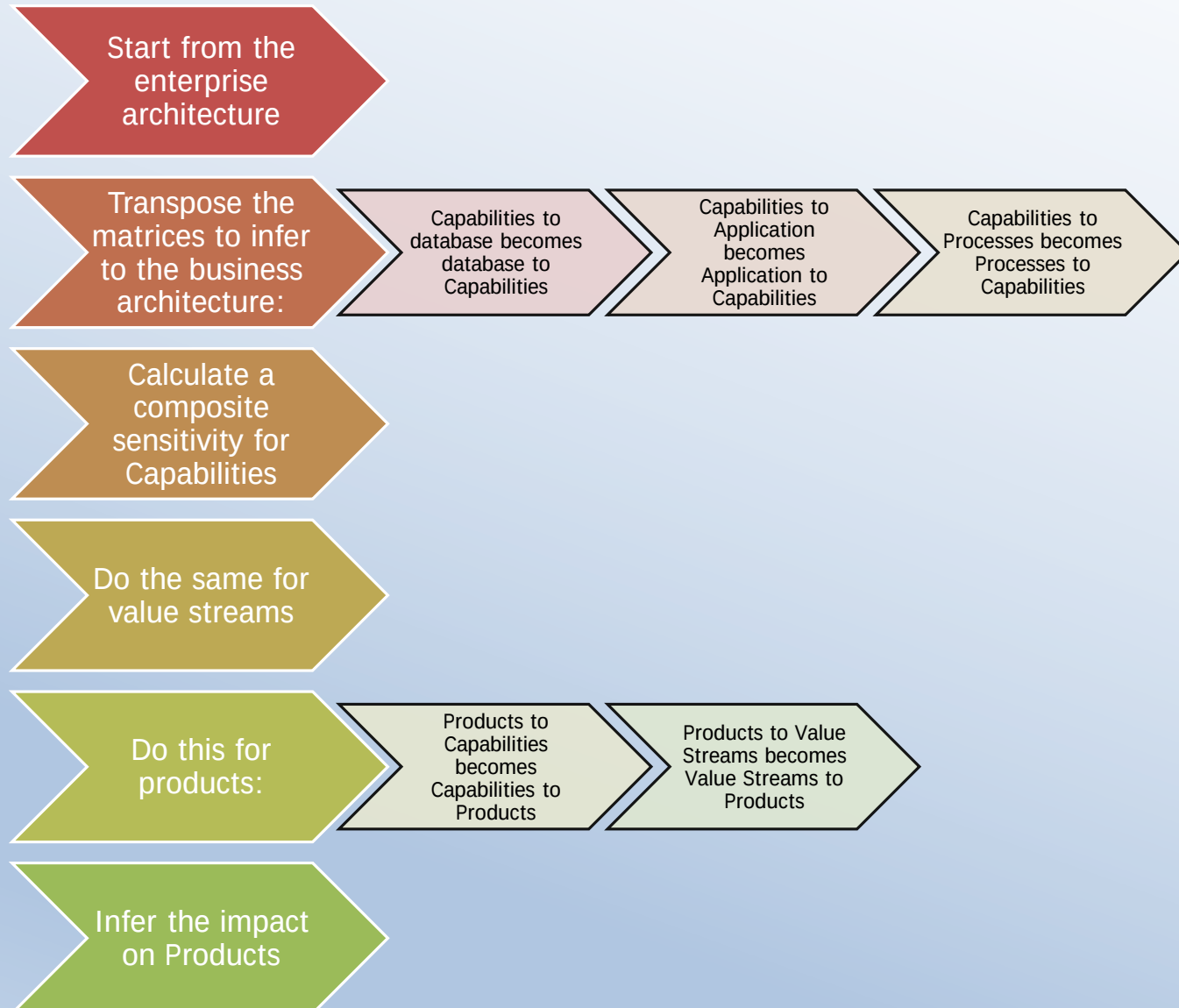
The business architecture is 'squeezed' from two directions

This requires inverse impact analysis from the Enterprise Architecture perspective along with Strategy to Business impact analysis

Inverse or reverse impact analysis

- Sometimes you want to assess the impact of a major enterprise architecture change such as:
 - Implementing a new ERP system
 - Shifting to cloud technology
 - Developing an extensive web presence
 - Going mobile
 - Adding AI to processes
- Changes like these impact the business architecture
- Assessing this requires reversing the impact analysis technique

How to reverse the technique



Remember that there are other analytics that are useful at each architecture perspective

Comparative analytics that deal with consolidation, merger and acquisition

Ranking analytics similar to what was used for impact to identify options

Best fit for matching attributes along with components of the architecture

Affinity analysis to group things

Predictive analytics with clustering to identify hidden relationships for deployment

Thank You!

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